

Daily Problem: 08–Feb–2026

Problem Statement

Let p be a polynomial with coefficients in $\mathbb{N}$ and $\deg(p) \geq 2$. Fix a nonempty alphabet Σ and define the language

$$L = \{ w \in \Sigma^* \mid |w| = p(n) \text{ for some } n \in \mathbb{N} \}.$$

Use the Myhill–Nerode theorem to show that L is not regular.

Proof

Pick a fixed symbol $a \in \Sigma$. Since membership in L depends only on length, it suffices to consider the unary sublanguage

$$L_a = L \cap \{a\}^* = \{ a^{p(n)} : n \in \mathbb{N} \}.$$

If L were regular, then L_a would also be regular (regular languages are closed under intersection with regular languages). So it is enough to show that L_a is not regular.

Myhill–Nerode setup

Define an equivalence relation $\equiv$ on $\{a\}^*$ by:

$$x \equiv y \iff \forall z \in \{a\}^*, xz \in L_a \iff yz \in L_a.$$

The Myhill–Nerode theorem states that L_a is regular if and only if $\equiv$ has finitely many equivalence classes. Thus, to prove L_a is not regular, it suffices to produce infinitely many pairwise $\equiv$ -inequivalent strings.

Key fact about polynomials of degree ≥ 2

Since p has degree at least 2 and has nonnegative coefficients, it is strictly increasing for all sufficiently large n , and its successive gaps grow without bound:

$$p(n+1) - p(n) \xrightarrow{n \rightarrow \infty} \infty.$$

In particular, for every integer $M \geq 1$ there exists N such that for all $n \geq N$,

$$p(n+1) - p(n) > M.$$

(For example, one may note that $p(n+1) - p(n)$ is a polynomial in n of degree $\deg(p) - 1 \geq 1$, hence unbounded.)

Constructing infinitely many distinguishable prefixes

For each $n \in \mathbb{N}$, let

$$x_n = a^{p(n)} \in L_a.$$

We will show that the strings $x_0, x_1, x_2, \dots$ are pairwise inequivalent under $\equiv$, i.e. for $m < n$,

$$x_m \not\equiv x_n.$$

Fix $m < n$. Consider the suffix

$$z = a^{p(m+1) - p(m)}.$$

Then

$$x_m z = a^{p(m)} a^{p(m+1) - p(m)} = a^{p(m+1)} \in L_a.$$

So $x_m z \in L_a$.

Now look at $x_n z$. Its length is

$$|x_n z| = p(n) + (p(m+1) - p(m)).$$

We claim this length is *not* equal to $p(t)$ for any $t \in \mathbb{N}$, provided n is taken sufficiently large relative to m . Indeed, choose n so large that

$$p(n+1) - p(n) > p(m+1) - p(m).$$

This is possible by the unbounded-gap property above.

Since p is increasing for large n , we have

$$p(n) < p(n) + (p(m+1) - p(m)) < p(n) + (p(n+1) - p(n)) = p(n+1).$$

Thus the length $|x_n z|$ lies strictly between $p(n)$ and $p(n+1)$. Because there is no polynomial value strictly between consecutive values $p(n)$ and $p(n+1)$, it follows that

$$|x_n z| \neq p(t) \quad \text{for all } t \in \mathbb{N},$$

and hence

$$x_n z \notin L_a.$$

Therefore we have found a distinguishing suffix z such that

$$x_m z \in L_a \quad \text{but} \quad x_n z \notin L_a,$$

which implies $x_m \neq x_n$.

Infinitely many equivalence classes

For each fixed m , we can choose infinitely many $n > m$ satisfying $p(n+1) - p(n) > p(m+1) - p(m)$, so x_m differs from infinitely many later x_n . In particular, the set $\{x_n : n \in \mathbb{N}\}$ contains infinitely many pairwise inequivalent strings, so $\equiv$ has infinitely many equivalence classes.

By the Myhill–Nerode theorem, L_a is not regular, and therefore L is not regular.

Conclusion

Since p has degree at least 2, the gaps $p(n+1) - p(n)$ are unbounded. This yields infinitely many Myhill–Nerode distinguishable prefixes $a^{p(n)}$, so the language $\{w : |w| = p(n) \text{ for some } n\}$ cannot be regular.